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**Department name:** Computer Science & Engineering

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**University name:** University of Chittagong.

**Course No.:** CSE 423

**Course Title:** Simulation & Modeling.

**Course Credits:** 3.00

**Assignment No.:** 01

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**Total Count of Students in this course/class/batch:** 61

**Total Count of Exercises (or questions or problems):** 197

**Chapter No. :** 2

**Chapter Name:** Simulation Examples

**The problem (question) serial number:** 37

**The problem (question) Description:** Construct the customer wise simulation table (sample as table 2.10 and 2.14 in textbook) with CustomerID, Interarrival time, Arrival-Time, When-Server-Available, Server-Chosen, Service-Time, Time-Service-Begins, Waiting-Time-in-Queue, Time-Service-Ends, Time-Customer-Spends-in-System, Server-Idle-Time for the customers in the queueing system in the table 3.2 in the textbook.

Answers: (37) We have to construct the customers wise simulation table for the customers in the queueing system in the table 3.2 in the text book.

Collecting the info about each customers from table 3.2 Future Event list.

| Customers | Arrival Time | Service Time | Service End |
|-----------|--------------|--------------|-------------|
| C1        | 0            | 4            | 4           |
| C2        | 1            | 3            | 6           |
| C3        | 2            | 2            | 8           |
| C4        | 8            | 3            | 11          |
| C5        | 11           | 4            | 15          |
| C6        | 18           | 5            | 23          |
| C7        | 23           | 4            | 27          |
| C8        | 25           | -            | -           |

For Customer 1  $\rightarrow$

$$\text{Arrival} = 0$$

$$\text{Servers Available} = 0$$

$$\text{Service begin} = 0$$

$$\text{Service End} = 4$$

$$\text{waiting} = 0 - 0 = 0$$

$$\text{Servers Idle} = 0$$

For Customer 2  $\rightarrow$

$$\text{Arrival} = 1$$

$$\text{Interarrival} = 1$$

$$\text{Servers available} = 4$$

$$(\text{end of C1} = 4)$$

$$\text{Service Begin} = 4$$

$$\text{Service End} = 6$$

$$\text{waiting} = 4 - 1 = 3$$

$$\text{Servers idle} = 0.$$

customer: 03 →

Arrival = 2

Interarrival = 1

Servers available = 6  
(c2 end at 6)

Service begin = 6

Service end = 11

waiting =  $11 - 6 - 2 = 4$

Servers idle = 0

customer: 05 →

Arrival = 11

Interarrival = 15

Servers available = 15

Servers begins = 15

Service ends = 16

waiting =  $15 - 11 = 4$

Servers idle = 0

customer: 07 →

Arrival = 23

Interarrival = 5

Servers arrival = 23

Service begin = 23

Service End = 27

waiting = 0

Servers idle = 0

customer: 04 →

Arrival = 8

interarrival = 6

Servers Available = 11  
(c3 end at 11)

Service begins = 11

Service ends = 15

waiting =  $11 - 8 = 3$

Servers idle = 0

customer: 06 →

Arrival = 18

Interarrival = 16

Service begins = 18

Service ends = 22

waiting = 0

Servers idle

$= 18 - 16 = 2$

customer: 08 →

Arrival = 25

Interarrival = 2

Servers Available = 27

Service begins = 27

waiting =  $27 - 25 = 2$

customers-wise simulation table.

| Customer ID | Interarrival Time | Arrival Time | When Server Available | Server Chosen | Service Time | Time Service Begins | Waiting Time in Queue | Time Service End | Time in system | Server idle Time |
|-------------|-------------------|--------------|-----------------------|---------------|--------------|---------------------|-----------------------|------------------|----------------|------------------|
| C1          | -                 | 0            | 0                     | single        | 4            | 0                   | 0                     | 4                | 4              | 0                |
| C2          | 1                 | 1            | 4                     | single        | 2            | 4                   | 3                     | 6                | 5              | 0                |
| C3          | 1                 | 2            | 6                     | single        | 5            | 6                   | 4                     | 11               | 9              | 0                |
| C4          | 6                 | 8            | 11                    | single        | 4            | 11                  | 3                     | 15               | 7              | 0                |
| C5          | 3                 | 11           | 15                    | single        | 1            | 15                  | 4                     | 16               | 5              | 0                |
| C6          | 7                 | 18           | 16                    | single        | 5            | 18                  | 0                     | 23               | 5              | 2                |
| C7          | 5                 | 23           | 23                    | single        | 4            | 23                  | 0                     | 27               | 4              | 0                |
| C8          | 2                 | 25           | 27                    | single        | -            | 27                  | 2                     | -                | -              | 0                |

Total customers (C1-C7):

14 - 39 2

single Servers Queue System.